

Reverse Proxy Setup for Therefore™ On-Premises

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1. Introduction

This White Paper demonstrates an example of configuring a reverse proxy based on IIS. A reverse proxy can also be set up with a multitude of different tools. Please consider your options carefully and select the optimal solution for your network.

2. Setting up a Reverse Proxy with Therefore™

2.1 Setup Requirements

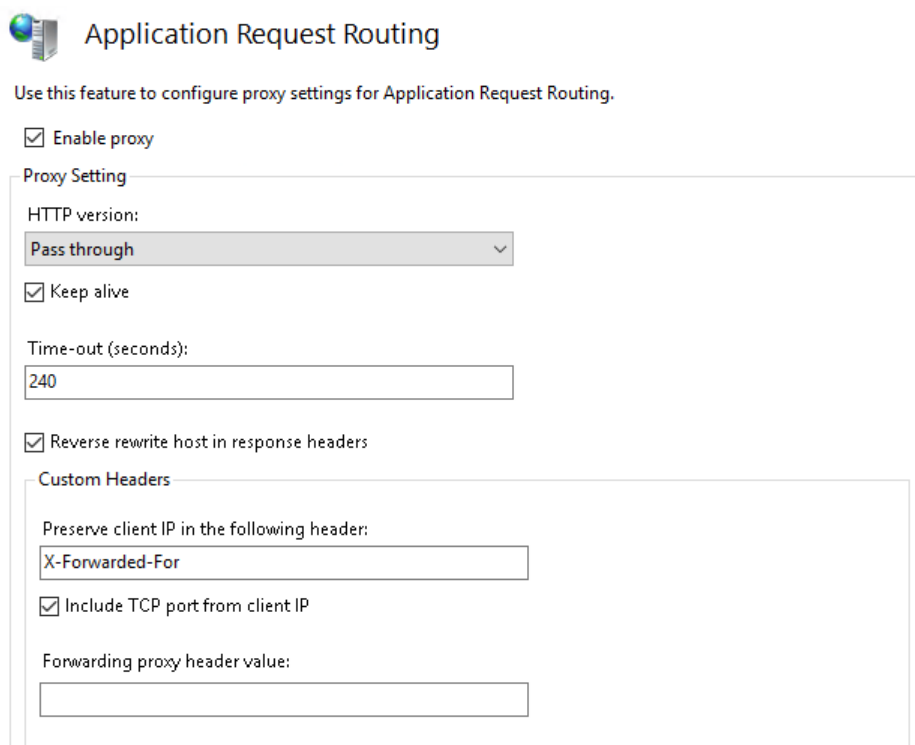
Please follow the steps described below in order to set up a reverse proxy with Therefore™ On-Premises.

For this solution, the following requirements must be met.

- Microsoft IIS must be installed on the Web Server
- A HTTPS connection must be configured on IIS with the proper certificate

2.2 Web Server Setup

Please follow the steps described below to complete the web server setup of your reverse proxy.



Application Request Routing

Use this feature to configure proxy settings for Application Request Routing.

☒ Enable proxy

Proxy Setting

HTTP version:
Pass through

☒ Keep alive

Time-out (seconds):
240

☒ Reverse rewrite host in response headers

Custom Headers

Preserve client IP in the following header:
X-Forwarded-For

☒ Include TCP port from client IP

Forwarding proxy header value:

1. Install: <https://www.iis.net/downloads/microsoft/url-rewrite>
2. Install <https://www.iis.net/downloads/microsoft/application-request-routing>
3. Enable the 'Application Request Routing' proxy:
 - a. Open IIS Manager
 - b. Select the Server Node in the tree view
 - c. Open 'Application Request Routing Cache'
 - d. In the 'Actions' pane click: 'Server Proxy Settings...'
 - e. Check the "Enable proxy" checkbox
 - f. Optional:

If you want to change the client IP header name fill in your custom header under '[Preserve client IP in the following header](#)'. For this option, please make sure that your custom header matches the name in the appsettings section of the XML server config file on the [app server](#).

g. In the Actions pane click: Apply.

4. Open your IIS default website route directory with File Explorer. Create the following two folders: 'TheXMLServer' and 'theservice'.

5. Inside of the new folder 'TheXMLServer' create a web.config file with the following content:

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
<system.webServer>
<rewrite>
<rules>
<clear />
<rule name="ReverseProxyXMLServer" stopProcessing="true">
<match url="(.*)" />
<conditions logicalGrouping="MatchAll" trackAllCaptures="false" />
<action type="Rewrite" url="https://enter your server name or the cluster role DNS name
here/TheXMLServer" />
</rule>
</rules>
</rewrite>
<httpProtocol>
<customHeaders>
<remove name="Content-type" />
</customHeaders>
</httpProtocol>
</system.webServer>
</configuration>
```

6. Inside of the new folder 'theservice' create a web.config file with the following content:

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
<system.webServer>
<rewrite>
<rules>
<clear />
<rule name="ReverseProxyWebAPI" stopProcessing="true">
<match url="(.*)" />
<conditions logicalGrouping="MatchAll" trackAllCaptures="false" />
<serverVariables>
<set name="REQUEST_SERVER" value="{HTTP_HOST}" />
</serverVariables>
<action type="Rewrite" url="https://enter your server name or the cluster role DNS name
here/theservice/{R:0}" />
</rule>
</rules>
<outboundRules>
<rule name="ChangeServerName" preCondition="ResponseIsHtml">
<match filterByTags="None" customTags="" pattern="(.*)://enter your server name or the
cluster role DNS name here/(.*)" />
```

```

<action type="Rewrite" value="{R:1}://{REQUEST_SERVER}/{R:2}" />
</rule>
<preConditions>
<preCondition name="ResponseIsHtml">
<add input="{RESPONSE_CONTENT_TYPE}" pattern="^text/html" />
</preCondition>
</preConditions>
<customTags>
</customTags>
</outboundRules>
</rewrite>
<httpProtocol>
<customHeaders>
<remove name="Content-type" />
</customHeaders>
</httpProtocol>
<security>
<requestFiltering>
<!-- ~ 4GB -->
<requestLimits maxAllowedContentLength="4294967295" />
</requestFiltering>
</security>
</system.webServer>
</configuration>

```

2.3 App Server Setup

1. Make sure the [Therefore™ XML Web Service](#) is installed on your app server.
2. Install your https certificate in Local Machine Personal store.
3. Install the certificate for the Therefore™ XML Web Service on port 443:
Run the following netsh command in an elevated Command Prompt:
netsh http add sslcert 0.0.0.0:443 certhash=[enter your certificate thumbprint here]
appid={89eb48e3-aa5d-4732-84ae-a83862d269d6}
4. Optional: Add firewall rule for port 443; For Windows firewalls using the Domain and Private profile use the following command:
New-NetFirewallRule -DisplayName "Therefore XML Server" -Direction Inbound -LocalPort 443 -Protocol TCP -Action Allow -Profile Private, Domain
5. Change the 'TheXMLServer.exe.config' file in the standard Therefore installation directory 'C:\Program Files\Therefore' by adding the following key in the 'appsettings' section: <add key="CustomIPHeader" value="X-Forwarded-For" /> Here, 'X-Forwarded-For' can be replaced by any value that can be configured in the reverse proxy. See also the [web server setup](#).